Title	**:** Script-Controlled ACL – Restrict Record Access Based on Field Value
College Name	**:** Balagan Saraswathi Arts and Science College for Women
College Code	**:** MSU144
University Name	**:** Manonmaniam Sundaranar University Tirunelveli
Course Name	**:** ServiceNow – System Admin with IT Service Management & AI Emphasis
Team ID	**:** SWTID-2026-3161
Team Leader	**:** Muthu Iswarya C
Team member	**:** a Sowmya S
Team member	**:** Dharshini M
Team member	**:** Esakiammal A

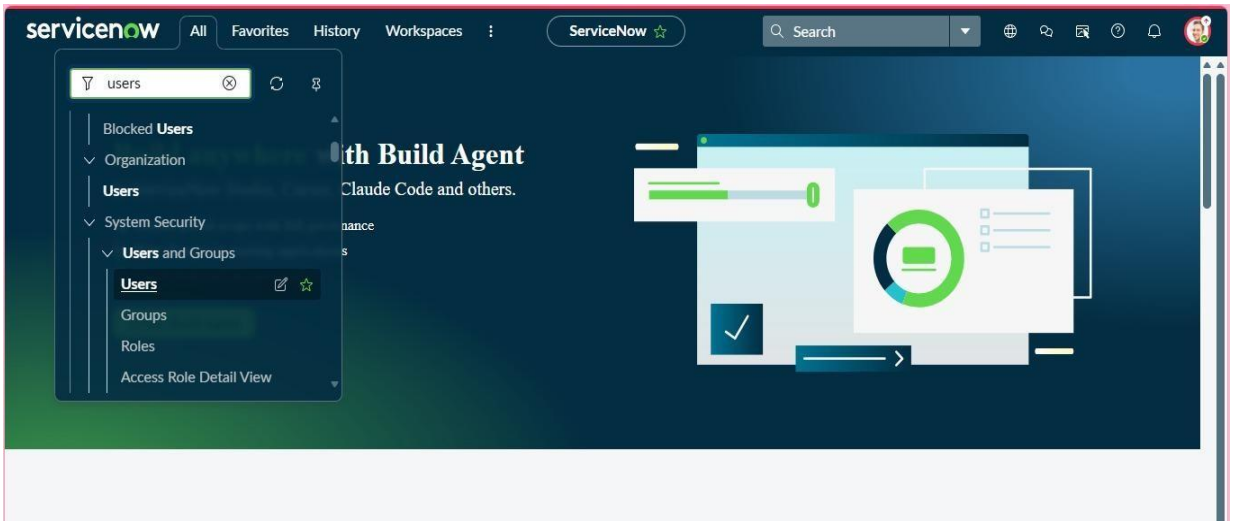
Script-Controlled ACL – Restrict Record Access Based on Field Value

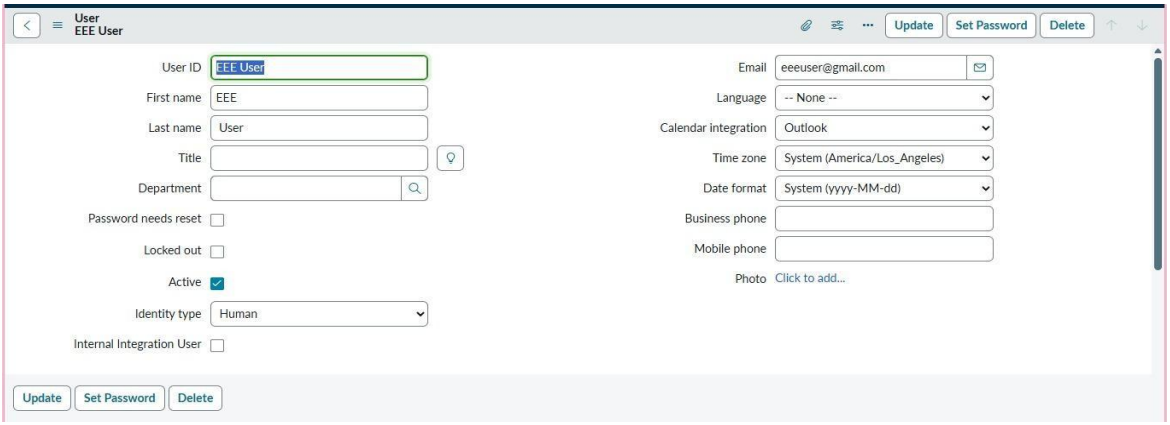
Introduction:

This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the **EEE branch** are allowed to view records where the **Branch field is set to EEE**, while administrators retain full access. **Milestone-1 Creation of Users and Roles**

User Creation:

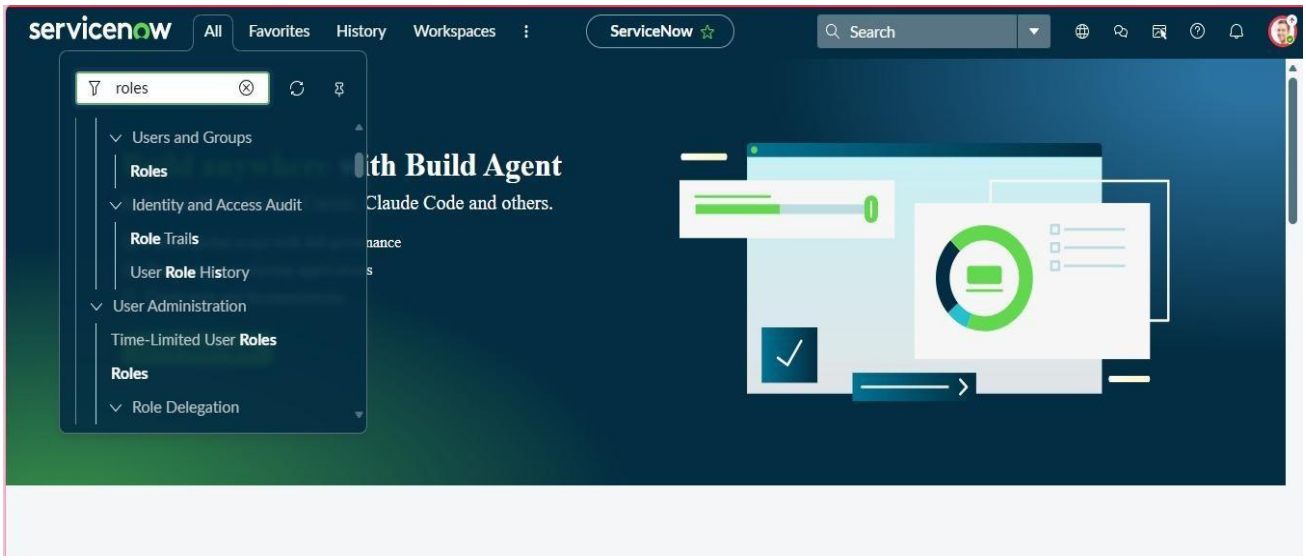
1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**
   - Navigate to **User Administration → Users**
   - Click **New** ○ **User ID**: EEE User ○ **First name**: EEE ○ **Last name**: User ○ **Email**: eeeuser@gmail.com ○ Save the record

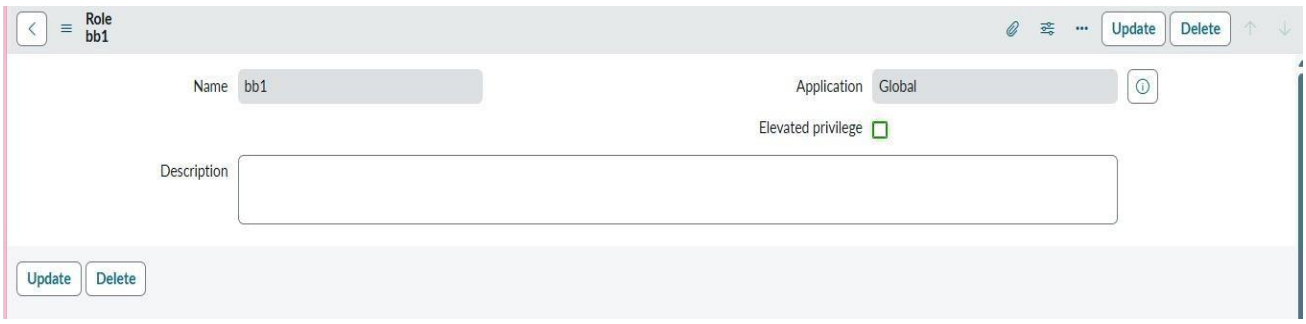




Role Creation

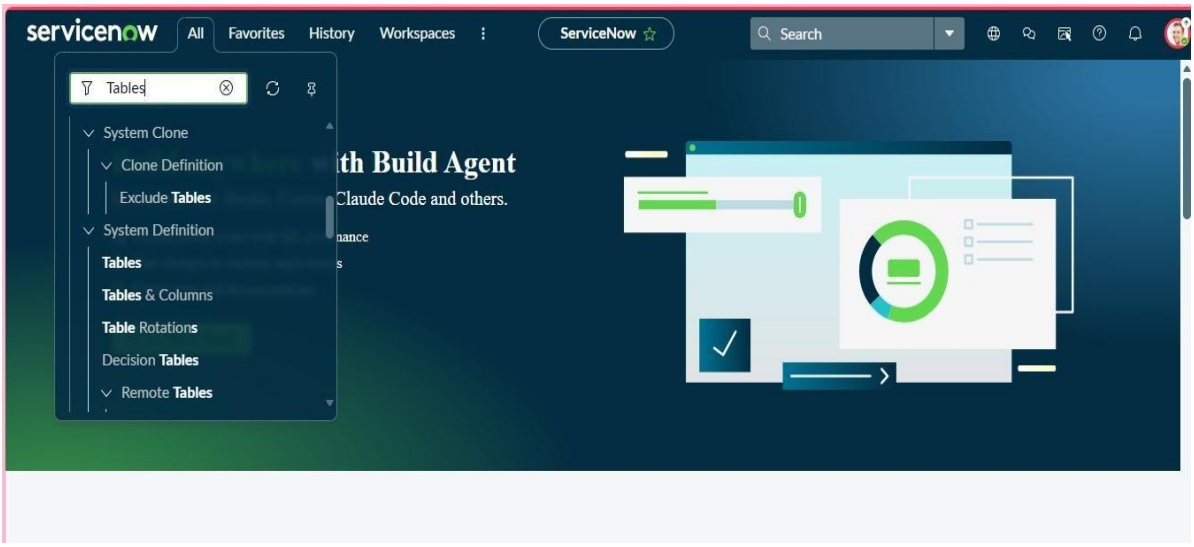
- Navigate to **User Administration → Roles**
- Click **New**
- **Name**: Enter role name as bb1
- **Description**: Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)** Assign all the roles to the EEE User

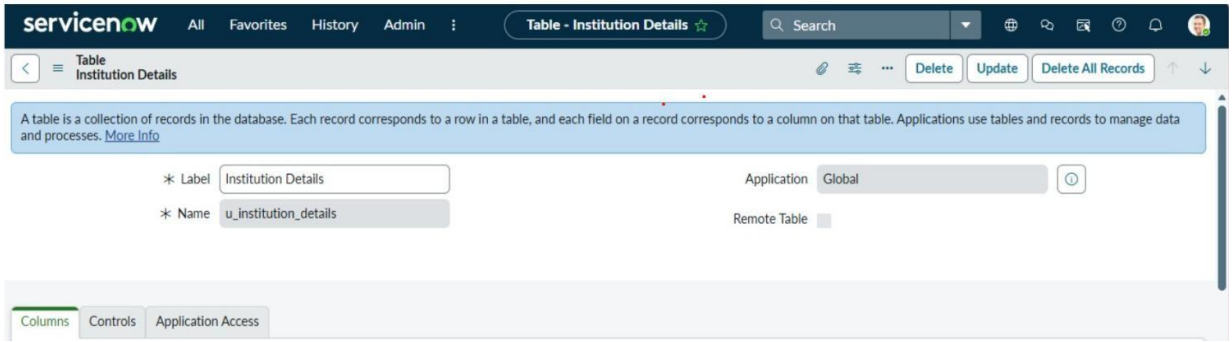




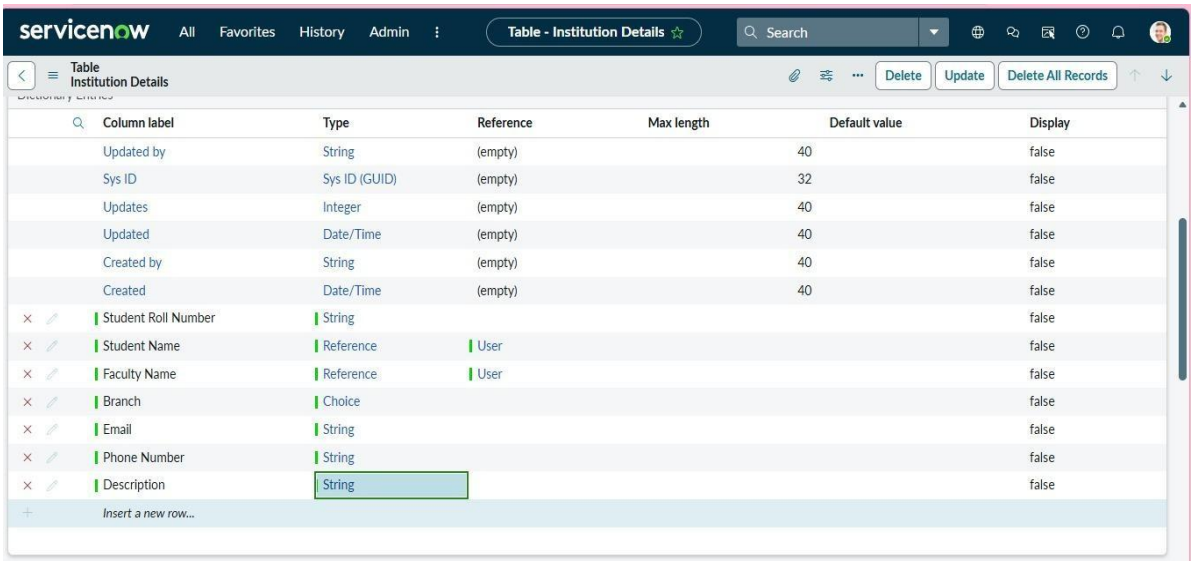
Tables Creation

1. In the **Application Navigator**, search for **System Definition → Tables**.
2. Click **New** and create a custom table with the following details:
    - **Label:** Institution Details
    - **Name:** u_institution_details
    - **Extends for :** false



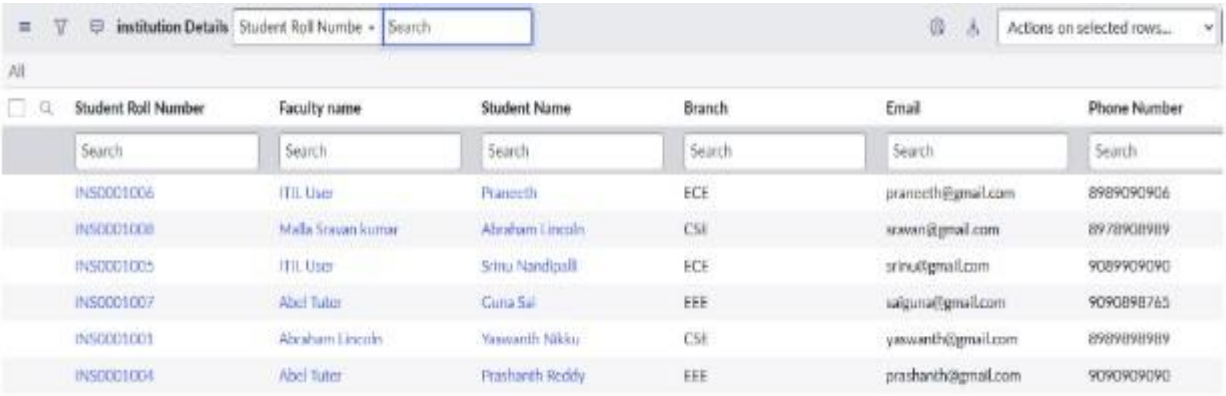


3. Click **Submit**.
4. Add the following fields to the **u_institution_details** table:
    - **Student Roll Number**– Auto Number
    - **Student Name** – Reference- User
    - **Faculty Name** – Reference- User
    - **Branch** – Choice (ECE, EEE, CSE)
    - **Email**– String
    - **Phone Number**– String
    - **Description**– Multi String
5. Save the Table



Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Updates	Integer	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Created by	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Student Roll Number	String				false
Student Name	Reference	User			false
Faculty Name	Reference	User			false
Branch	Choice				false
Email	String				false
Phone Number	String				false
Description	String				false
Insert a new row...					

6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).



Student Roll Number	Faculty name	Student Name	Branch	Email	Phone Number
Search	Search	Search	Search	Search	Search
INS0001006	ITIL User	Praneeth	ECE	praneeth@gmail.com	8989090906
INS0001008	Malla Sravan kumar	Abraham Lincoln	CSE	sravan@gmail.com	8978989889
INS0001005	ITIL User	Srinu Nandipalli	ECE	srinu@gmail.com	9089909090
INS0001007	Abel Tuter	Guna Sai	EEE	saiguna@gmail.com	9090898765
INS0001001	Abraham Lincoln	Yaswanth Nikku	CSE	yaswanth@gmail.com	8989898989
INS0001004	Abel Tuter	Prashanth Reddy	EEE	prashanth@gmail.com	9090909090

Creation of Access Control Lists(Read ACL)

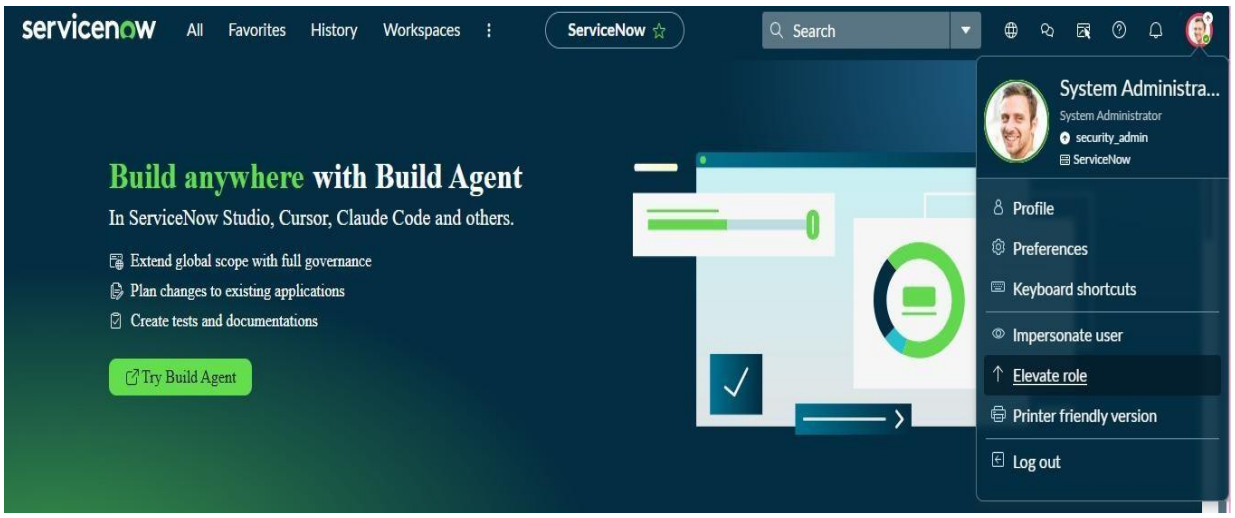
1. In the **Application Navigator**, search for **System Security → Access Control (ACL)**.
2. Elevate “Security_admin” role 3. Click **New** to create a new ACL.
4. Enter the following details:
   - **Type:** record ○ **Operation:** read ○ **Name:** u_institution_details ○ **Active:** true ○ **Advanced:** true
5. In the **Requires role** related list, add:
   - bb1 (custom role)
6. Add Data condition as **Branch is EEE**

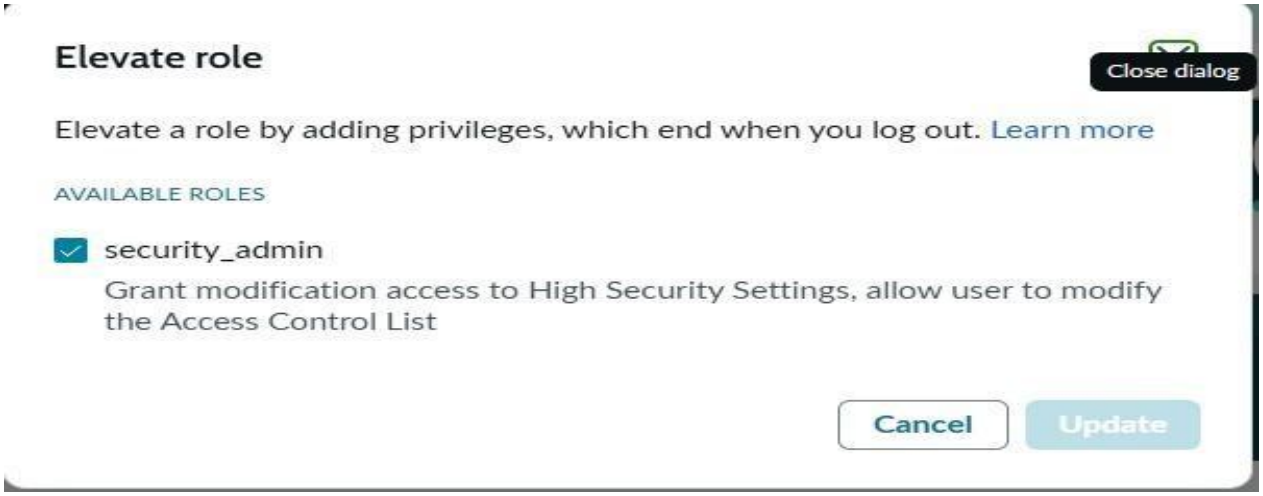
7. In the **Script** field, add the following script:

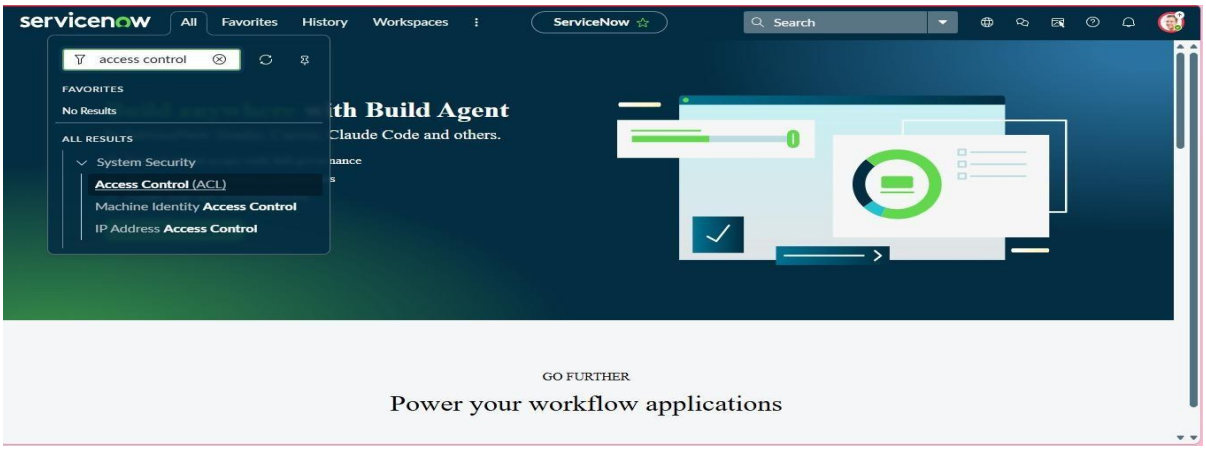
```
(function () {

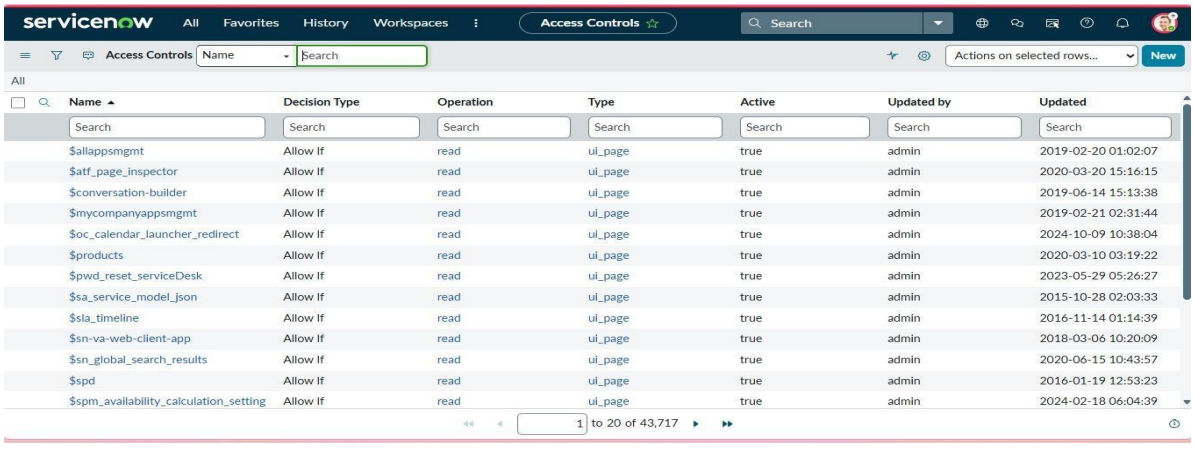
   // Allow admin users full access
if (gs.hasRole('admin')) {
      return true;
   }
   // Allow only EEE branch users to see EEE records
if (gs.hasRole('bb1')){
      return true;
   }
   // Deny access for all others
   return false;
})();
```

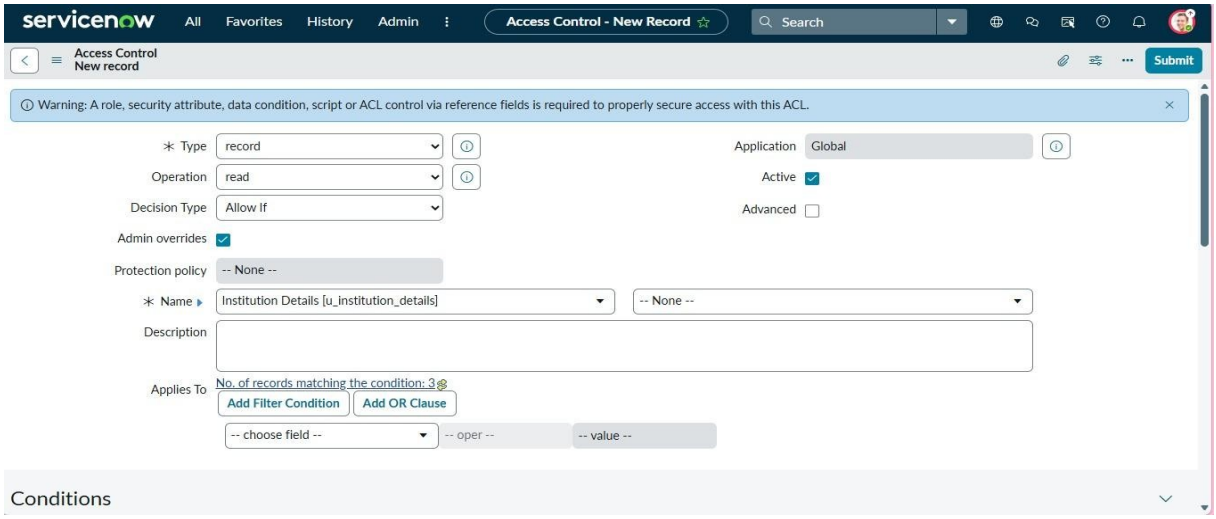
8. Click **Submit** to save the ACL.

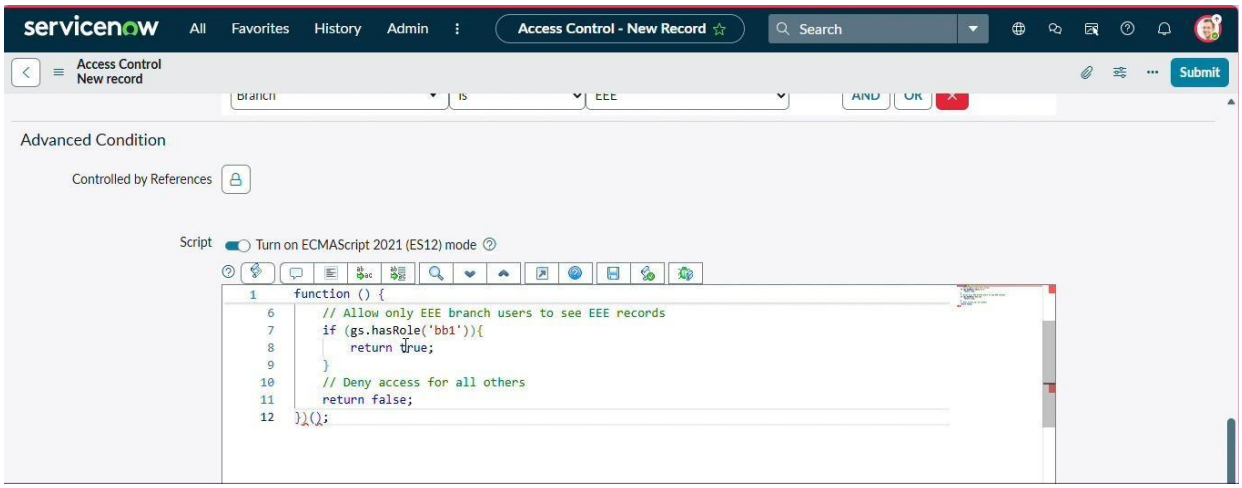



Elevate role
Close dialog
Elevate a role by adding privileges, which end when you log out. Learn more
AVAILABLE ROLES
security_admin
Grant modification access to High Security Settings, allow user to modify the Access Control List
Cancel
Update


servicenow
All
Favorites
History
Workspaces
ServiceNow
Search
access control
FAVORITES
No Results
ALL RESULTS
System Security
Access Control (ACL)
Machine Identity Access Control
IP Address Access Control
th Build Agent
Claude Code and others.
GO FURTHER
Power your workflow applications

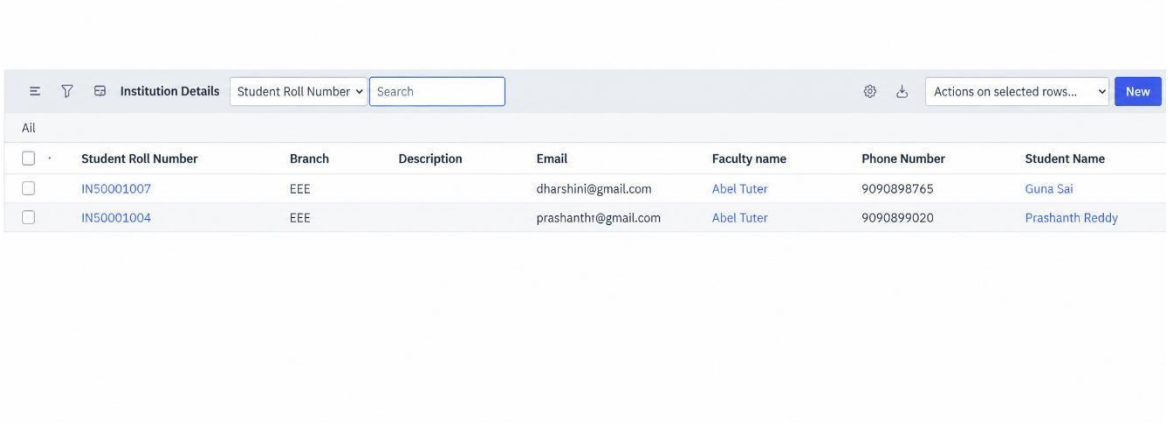

servicenow
All
Favorites
History
Workspaces
Access Controls
Search
Access Controls
Name
Search
Actions on selected rows...
New
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
$allappsmgmt Allow If read ui_page true admin 2019-02-20 01:02:07
$atf_page_inspector Allow If read ui_page true admin 2020-03-20 15:16:15
$conversation-builder Allow If read ui_page true admin 2019-06-14 15:13:38
$mycompanyappsmgmt Allow If read ui_page true admin 2019-02-21 02:31:44
$oc_calendar_launcher_redirect Allow If read ui_page true admin 2024-10-09 10:38:04
$products Allow If read ui_page true admin 2020-03-10 03:19:22
$pwd_reset_serviceDesk Allow If read ui_page true admin 2023-05-29 05:26:27
$sa_service_model_json Allow If read ui_page true admin 2015-10-28 02:03:33
$sla_timeline Allow If read ui_page true admin 2016-11-14 01:14:39
$sn-va-web-client-app Allow If read ui_page true admin 2018-03-06 10:20:09
$sn_global_search_results Allow If read ui_page true admin 2020-06-15 10:43:57
$spd Allow If read ui_page true admin 2016-01-19 12:53:23
$spm_availability_calculation_setting Allow If read ui_page true admin 2024-02-18 06:04:39
1 to 20 of 43,717





9. Impersonate **EEE users** and open the Student Records list.
10. Copy the table name u_institution_details.list/LIST and paste in the All menu.
11. Impersonate **Admin** and open the Student Records list. **Verify :**

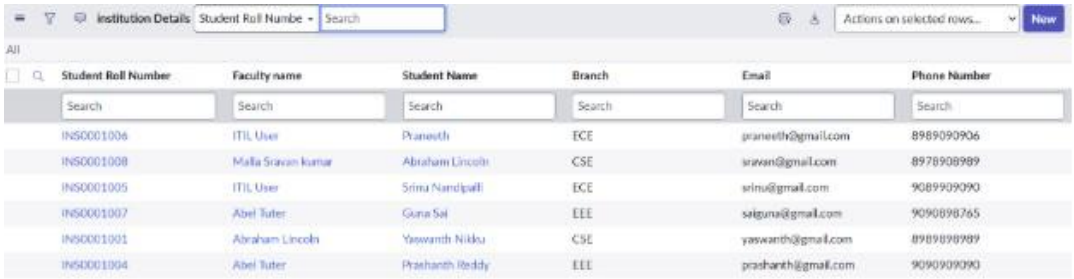
- User with **bb1 role** can view **only EEE branch records**



- User without the role cannot view any records



- Admin user can view **all records regardless of branch**



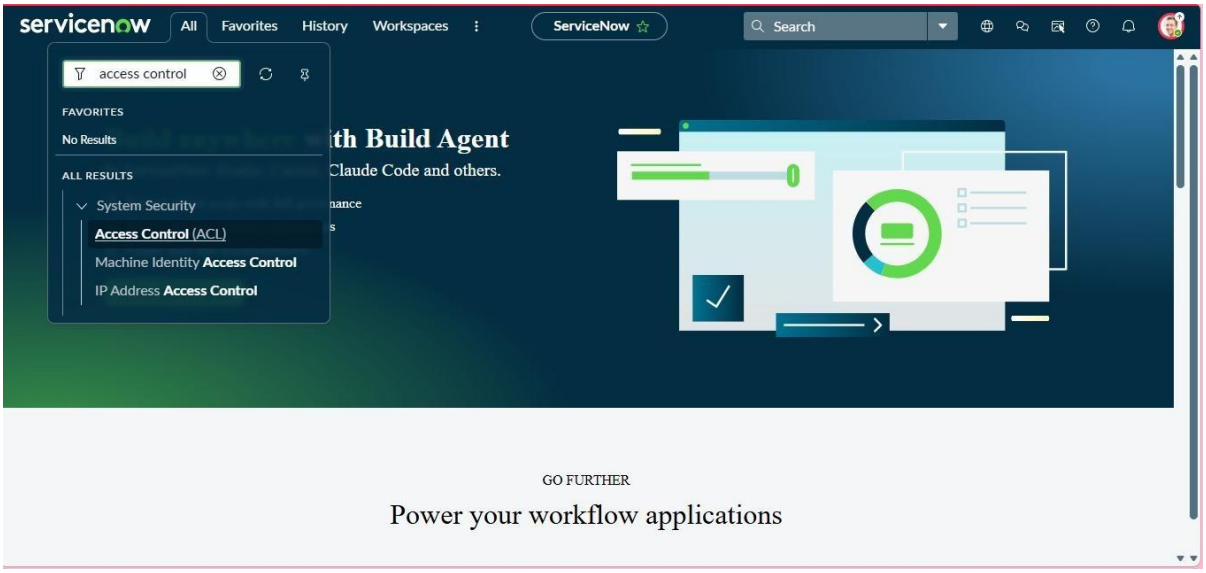
Creation of Access Control Lists(Create ACL)

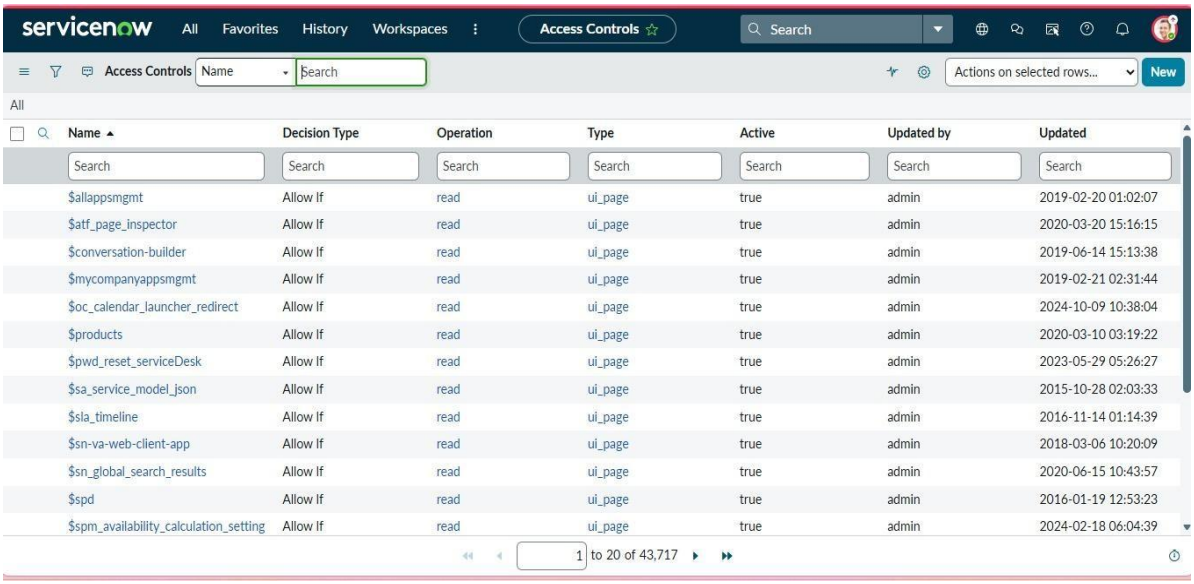
1. Click **New** to create a new ACL(Create) 2.
Enter the following details:

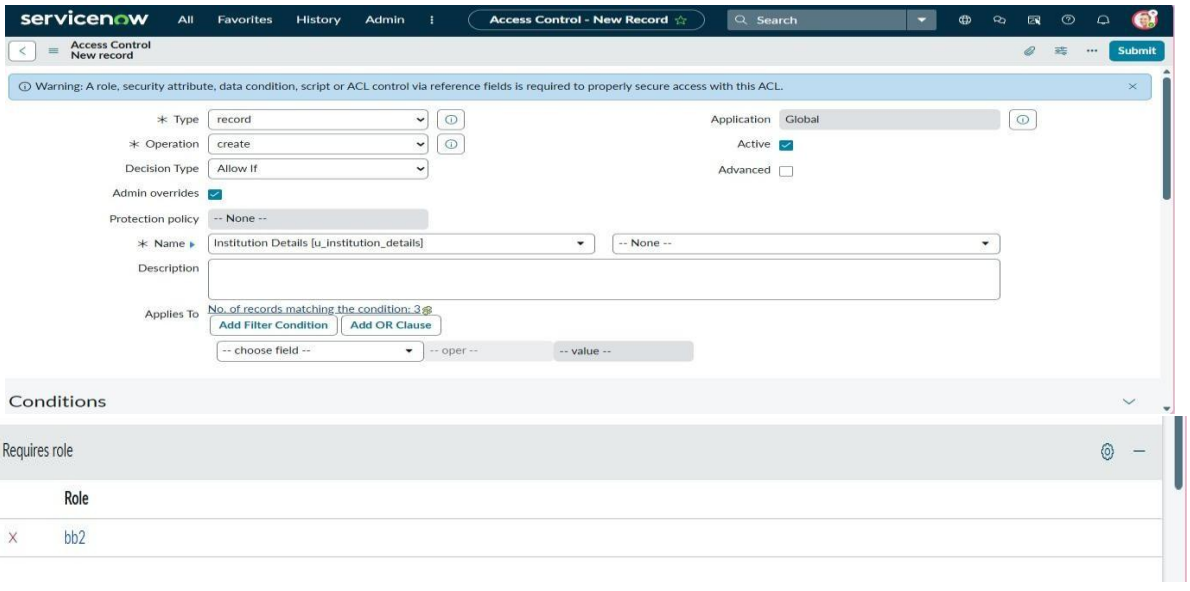
- **Type:** record
- **Operation:** create
- **Name:**

u_institution_details
- **Active:** true

3. In the **Requires role** related list, add:
   - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record

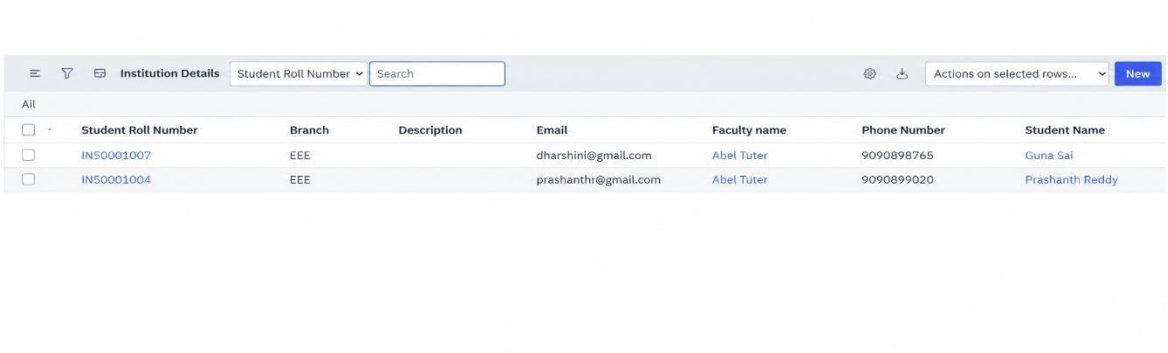

servicenow
All
Favorites
History
Workspaces
ServiceNow
Search
access control
FAVORITES
No Results
ALL RESULTS
System Security
Access Control (ACL)
Machine Identity Access Control
IP Address Access Control
th Build Agent
Claude Code and others.
nance
s
GO FURTHER
Power your workflow applications


servicenow
All
Favorites
History
Workspaces
Access Controls
Search
Access Controls
Name
Search
Actions on selected rows...
New
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
$allappsmgmt Allow If read ui_page true admin 2019-02-20 01:02:07
$atf_page_inspector Allow If read ui_page true admin 2020-03-20 15:16:15
$conversation-builder Allow If read ui_page true admin 2019-06-14 15:13:38
$mycompanyappsmgmt Allow If read ui_page true admin 2019-02-21 02:31:44
$oc_calendar_launcher_redirect Allow If read ui_page true admin 2024-10-09 10:38:04
$products Allow If read ui_page true admin 2020-03-10 03:19:22
$pwd_reset_serviceDesk Allow If read ui_page true admin 2023-05-29 05:26:27
$sa_service_model_json Allow If read ui_page true admin 2015-10-28 02:03:33
$sla_timeline Allow If read ui_page true admin 2016-11-14 01:14:39
$sn-va-web-client-app Allow If read ui_page true admin 2018-03-06 10:20:09
$sn_global_search_results Allow If read ui_page true admin 2020-06-15 10:43:57
$spd Allow If read ui_page true admin 2016-01-19 12:53:23
$spm_availability_calculation_setting Allow If read ui_page true admin 2024-02-18 06:04:39
1 to 20 of 43,717


servicenow
All
Favorites
History
Admin
Access Control - New Record
Search
Access Control
New record
Submit
Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.
Type record
Operation create
Decision Type Allow If
Admin overrides
Protection policy -- None --
Name Institution Details [u_institution_details]
Description
Applies To
No. of records matching the condition: 3
Add Filter Condition
Add OR Clause
-- choose field --
-- oper --
-- value --
Application Global
Active
Advanced
-- None --
Conditions
Requires role
Role
bb2

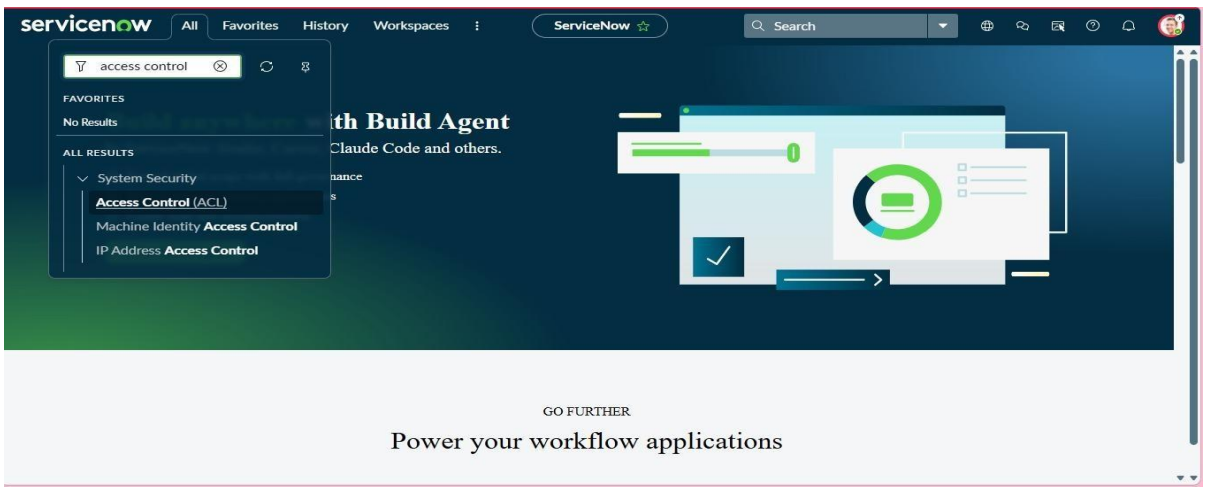
Verify :

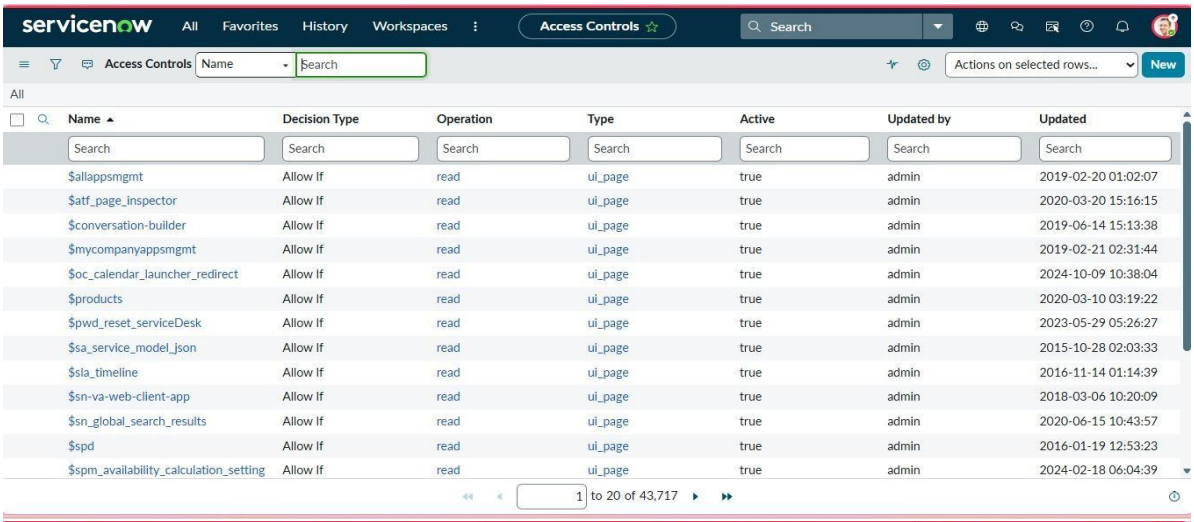
- User with **bb1 &bb2 roles** can view **only EEE branch records b New** Button

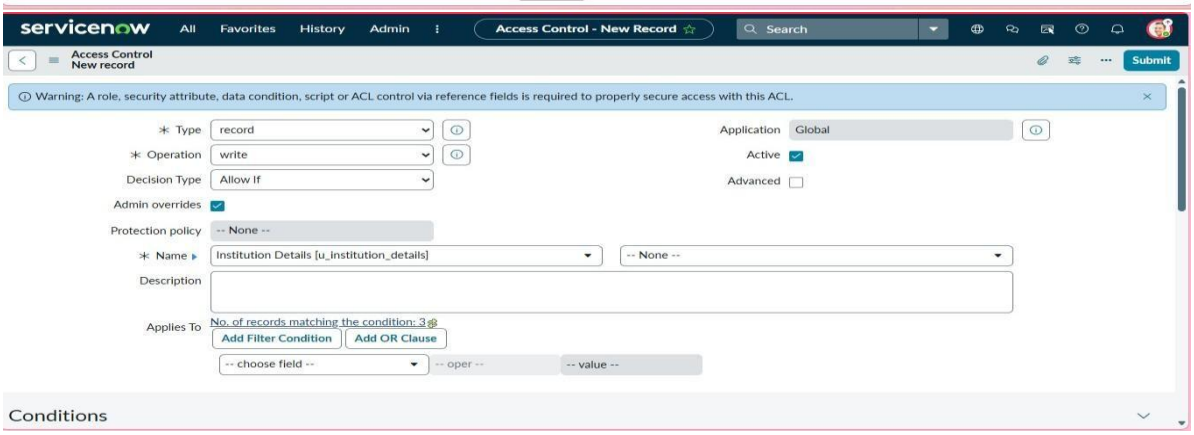


Creation of Access Control Lists(Write ACL)

1. Click New to create a new ACL(Write) 2.
Enter the following details:
    - Type: record ○
      Operation: Write ○
      Name: u_institution_details
    - Active: true
3. In the Requires role related list, add:
    - bb3 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



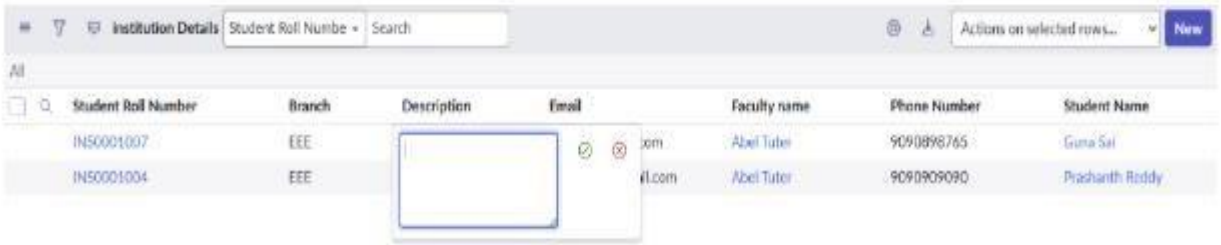


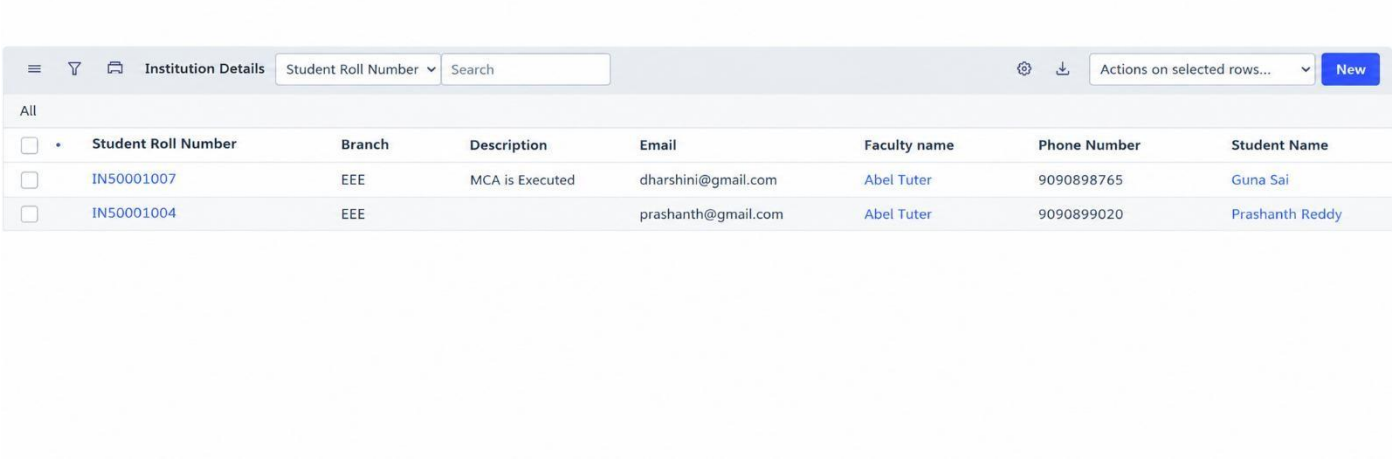




Verify :

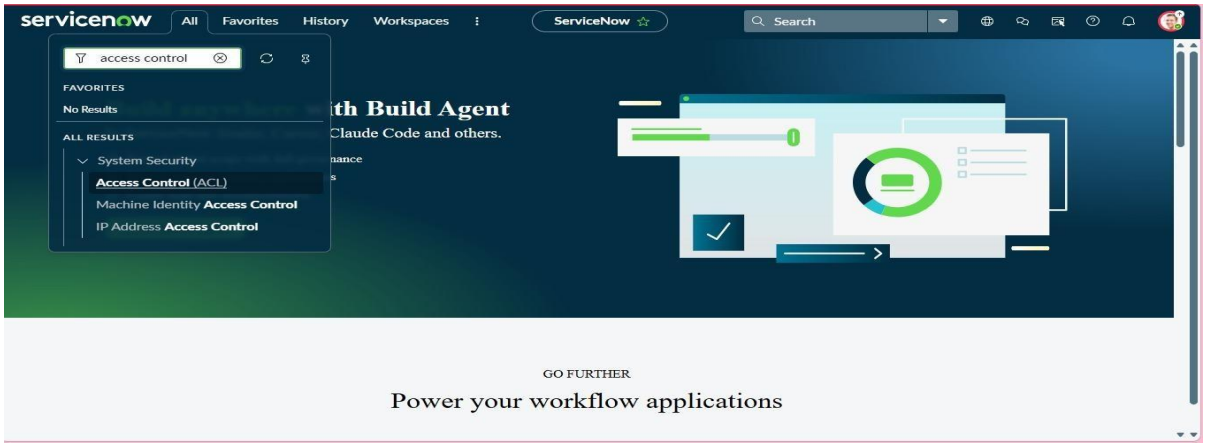
- User with **bb1 &bb2 &bb3 roles** can view and edit the **EEE branch records along with New** Button

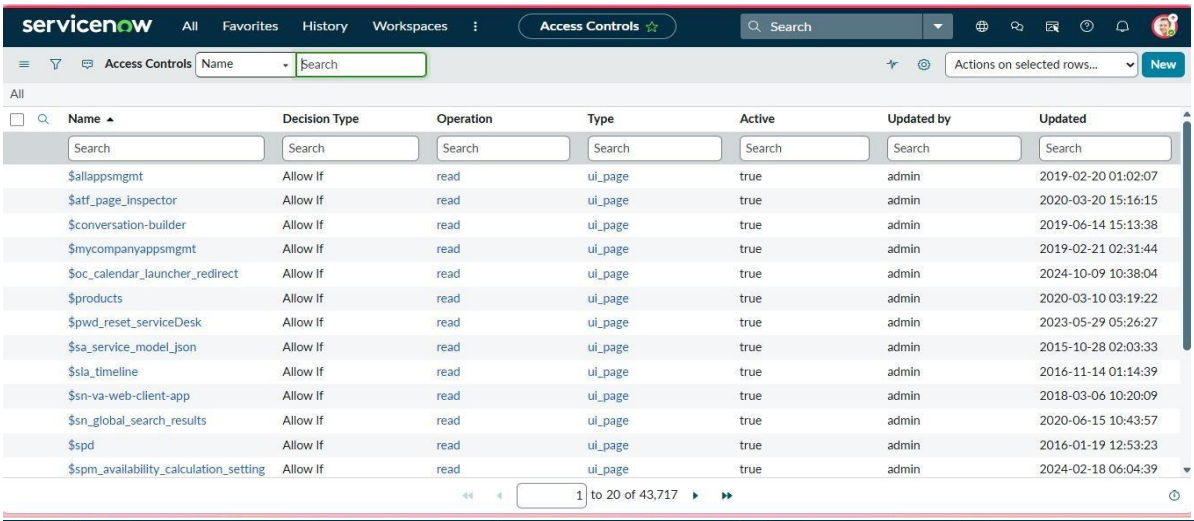


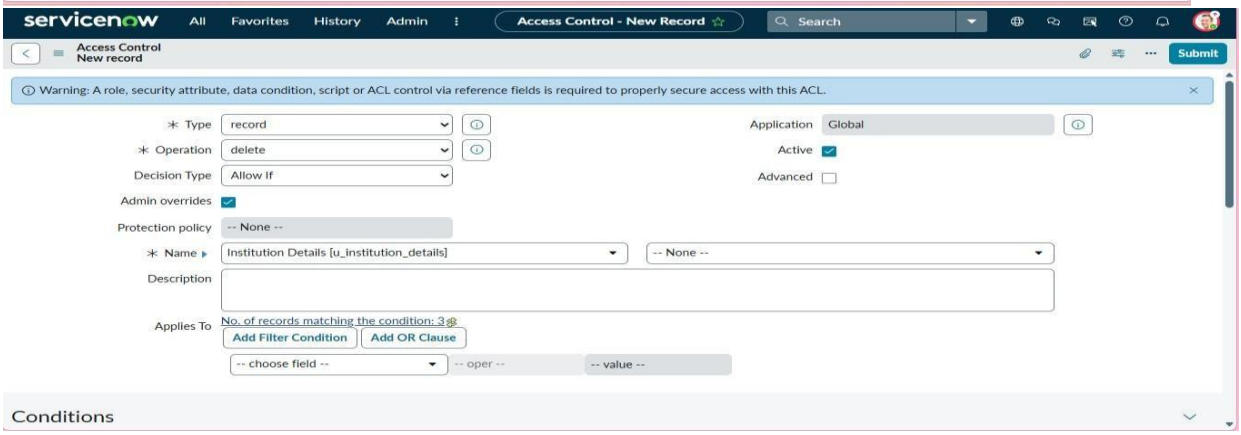


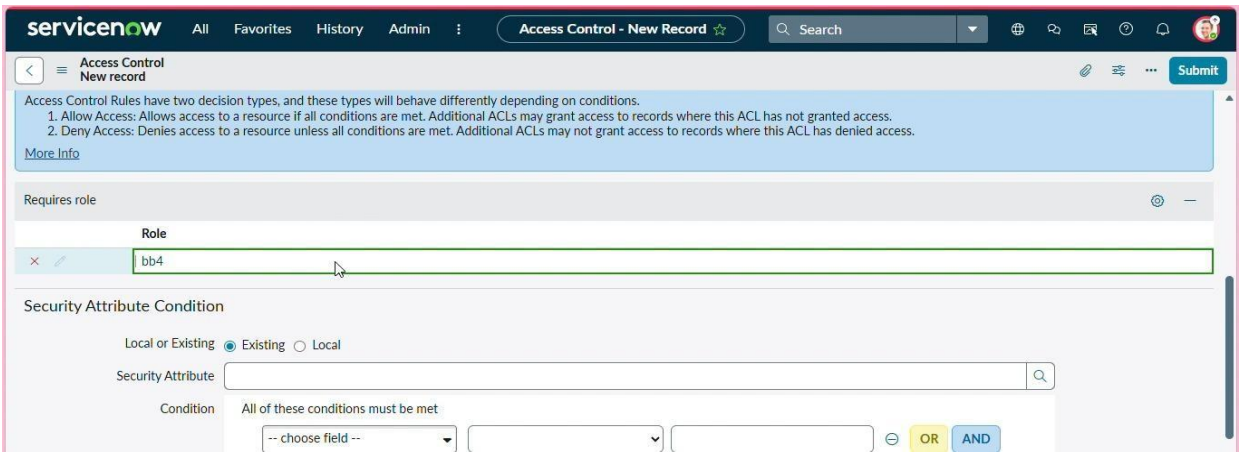
Creation of Access Control Lists(Delete ACL)

1. Click **New** to create a new ACL(Delete)
2. Enter the following details: ○ **Type:** record ○ **Operation:** Delete ○ **Name:** u_institution_details ○ **Active:** true
3. In the **Requires role** related list, add:
    - ○ bb4 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



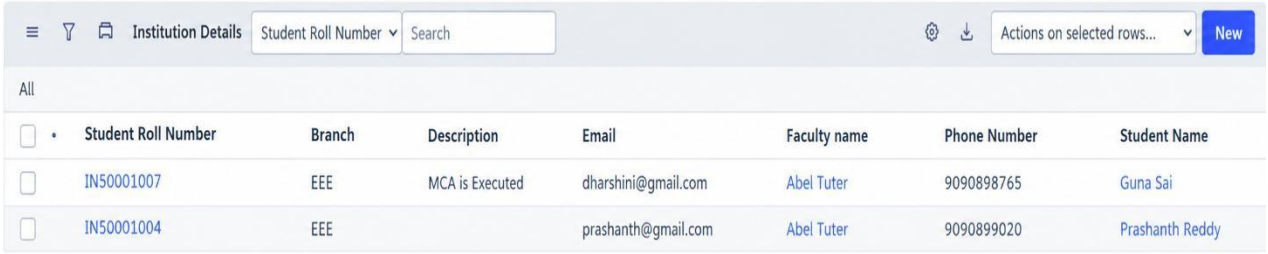


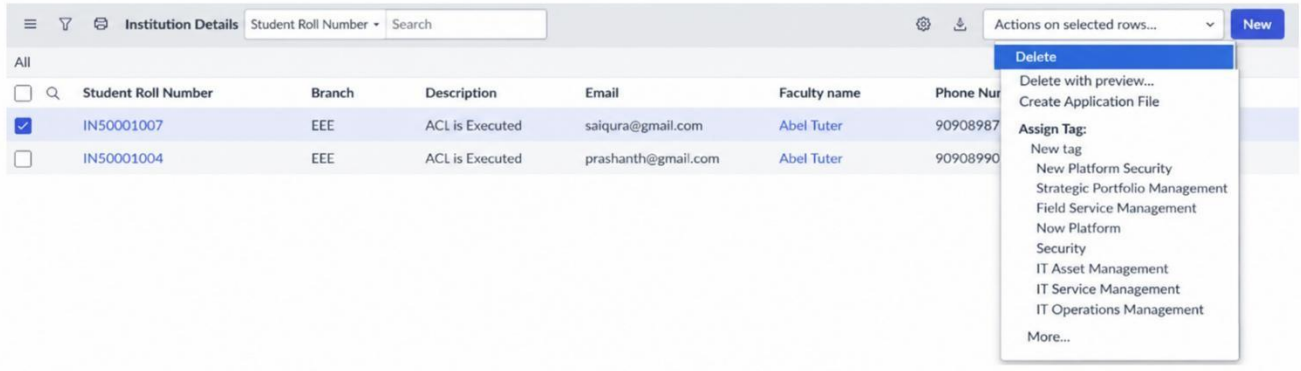


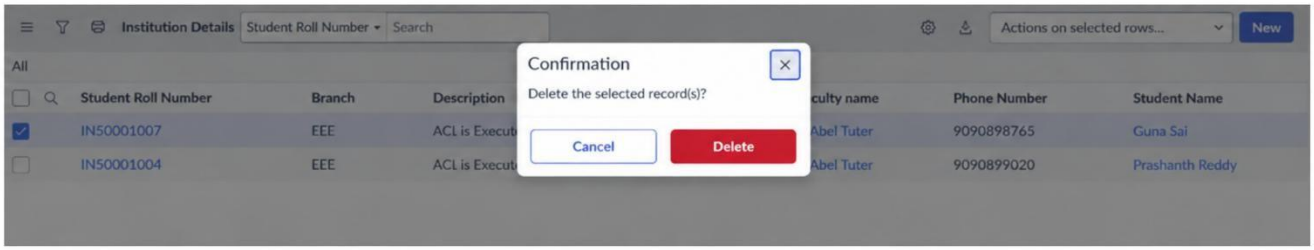


Verify :

- Users with **bb1 &bb2 &bb3&bb4 roles** can see, create and edit along with delete the **EEE branch records.**







Outcome :

By completing this lab, learners gain a comprehensive understanding of how scriptcontrolled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.

Advantages

- Provides secure record-level access in ServiceNow.

- Allows access based on user roles and field values.
- Protects sensitive student information from unauthorized users.
- Supports READ, CREATE, WRITE, and DELETE access control.
- Administrators can retain full access to all records.
- Helps prevent unauthorized viewing, editing, and deletion of records.

Disadvantages

- Creating and managing multiple ACLs can be complex.
- Incorrect scripts or role settings may prevent authorized users from accessing records.
- Requires proper knowledge of ServiceNow security and ACL configuration.
- Managing many custom roles can increase administration work.
- Changes in branch or user requirements may require updating the ACL configuration.

Future Enhancement

In the future, the Script-Controlled ACL system can be enhanced with more flexible and advanced access control features. Access permissions can be extended to different branches, departments, and user categories. Dynamic conditions can be introduced based on fields such as department, year, designation, or user role. The system can also include detailed access logs and monitoring features to track user activities. Automated testing can be implemented to verify ACL permissions whenever changes are made. In addition, the ACL system can be integrated with other ServiceNow applications and workflows to provide centralized and efficient security management. These enhancements can make the system more flexible, scalable, and easier to manage.

Conclusion

The **Script-Controlled ACL** provides an effective way to control access to records in ServiceNow. It combines user roles and field values to determine whether a user can view, create, modify, or delete records. In this project, access to EEE branch records is controlled using different roles and ACLs. This helps protect sensitive information and prevents unauthorized access. Overall, the project provides a practical understanding of record-level security and demonstrates how ACLs can be used to improve data protection in ServiceNow.